

Outline

Motivation: The Integrals We Cannot Solve

The Monte Carlo Principle

From Integral to Expectation

Crude Monte Carlo: Algorithm

Worked Toy Examples

Law of Large Numbers and Monte Carlo Error

Crude Monte Carlo in Practice

Importance Sampling

Variance Reduction Ideas

Practical Monte Carlo Diagnostics

Unit Summary

Rewriting an Integral as an Expectation

Any integral of the form

$$\theta = \int_{\mathcal{X}} g(x) dx$$

can be rewritten by introducing any density f with $f(x) > 0$ wherever $g(x) \neq 0$:

$$\theta = \int_{\mathcal{X}} \frac{g(x)}{f(x)} f(x) dx = E_f \left[\frac{g(X)}{f(X)} \right], \quad X \sim f.$$

$\mathcal{X} = \mathbb{R}^d$

$$\int g(x) dx$$

$$= \int \frac{g(x)}{f(x)} f(x) dx$$

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$$= E_f \left[\frac{g(X)}{f(X)} \right], X \sim f$$

$$\Rightarrow X_1, \dots, X_n \sim f$$

$$= \frac{1}{n} \sum_{i=1}^n \frac{g(x_i)}{f(x_i)}$$

$$\approx \int g(x) dx$$

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Special case: f already a density

If $\theta = \int \underline{h(x)f(x)} dx$ with $\underline{f}$ a genuine probability density, simply take $g(x)/f(x) = h(x)$:

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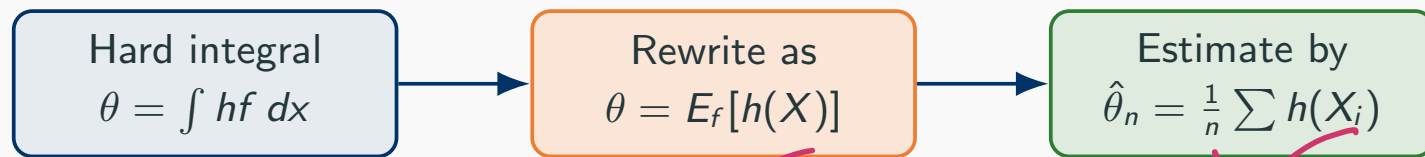
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This is the **entire trick** of Monte Carlo: turn “integrate” into “take an expectation”, then estimate the expectation with a sample average.

The Monte Carlo Principle: Integral $\rightarrow$ Expectation $\rightarrow$ Average

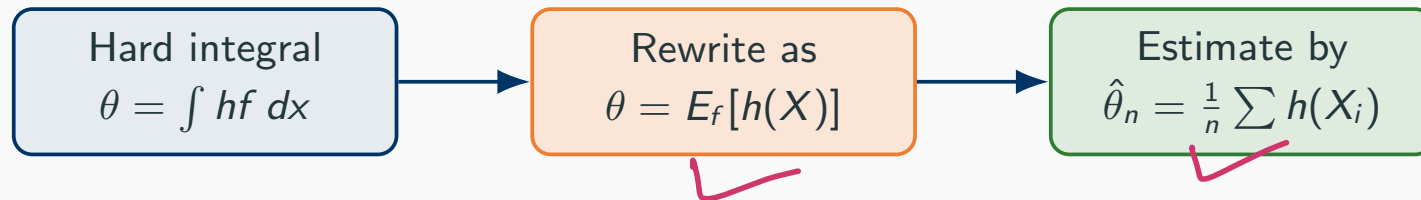


$\Rightarrow$ Ex: $\int_{-\infty}^{\infty} e^{-x^2} dx \rightarrow$ Explicit $\sqrt{\pi}$

$\Rightarrow$ $\int_{-\infty}^{\infty} \frac{e^{-x^2}}{\frac{1}{\sqrt{2\pi}} e^{x^2/2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \simeq \frac{1}{n} \sum_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-x_i^2/2}$
 $\rightarrow X_1, \dots, X_n \sim \mathcal{N}(0, 1)$

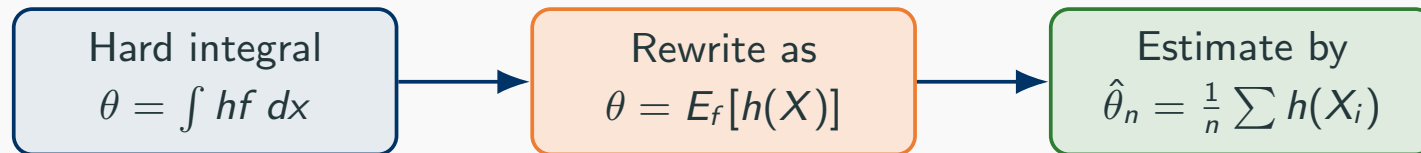
$\simeq \int_{-\infty}^{\infty} \frac{e^{-x^2}}{\frac{1}{\pi(1+x^2)}} dx \simeq \frac{1}{n} \sum_{i=1}^n \pi(1+x_i^2) e^{-x_i^2}$

The Monte Carlo Principle: Integral $\rightarrow$ Expectation $\rightarrow$ Average



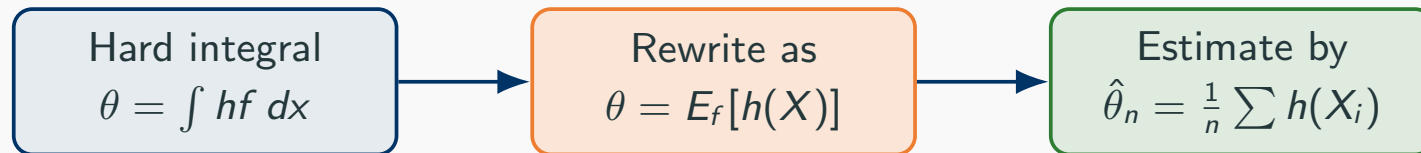
- Step 1 (integral $\rightarrow$ expectation) is pure algebra – always possible.

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- Step 1 (integral $\rightarrow$ expectation) is pure algebra – always possible.
- Step 2 (expectation $\rightarrow$ average) requires **Module 1**: we must be able to draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$.

The Monte Carlo Principle: Integral $\rightarrow$ Expectation $\rightarrow$ Average



- Step 1 (integral $\rightarrow$ expectation) is pure algebra – always possible.
- Step 2 (expectation $\rightarrow$ average) requires **Module 1**: we must be able to draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$.
- Everything in Module 2 is about making Step 2 as accurate and efficient as possible.

Crude (Basic) Monte Carlo Estimator

Algorithm: Crude Monte Carlo

1. Generate $\underline{X_1}, \underline{X_2}, \dots, \underline{X_n} \stackrel{\text{iid}}{\sim} f$.
2. Compute $\underline{h(X_1)}, \dots, \underline{h(X_n)}$.
3. Return

$$\theta = \int g(x) dx$$

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i)$$

$$, X_i \sim f$$

$$\Rightarrow E_f[\theta_n] \approx \theta$$

$$E_f \left[\frac{1}{n} \sum_{i=1}^n h(X_i) \right] = \frac{1}{n} \cdot \sum_{i=1}^n E_f(h(X_i)) = \frac{1}{n} \sum_{i=1}^n \int g(x_i) dx_i$$
$$= \frac{1}{n} \sum_{i=1}^n \int g(x) f(x) dx = \int g(x) f(x) dx = \int \frac{g(x)}{f(x)} \cdot f(x) dx = \int g(x) dx = \theta$$

$$E_f[h(X)] = \int h(x) f(x) dx = \int \frac{g(x)}{f(x)} \cdot f(x) dx = \int g(x) dx$$

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Unbiasedness

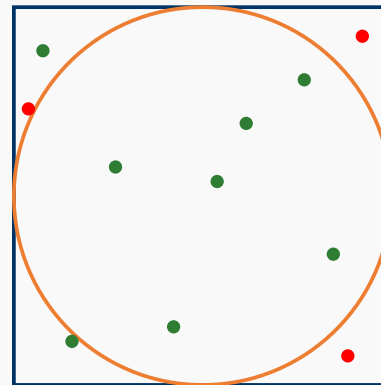
$$E[\hat{\theta}_n] = \frac{1}{n} \sum_{i=1}^n E[h(X_i)] = \frac{1}{n} \cdot n \cdot \underbrace{E_f[h(X)]}_{\checkmark} = \theta.$$

So $\hat{\theta}_n$ is an unbiased estimator of θ , for any sample size $n \geq 1$.

Classic Example: Estimating π by Throwing Darts

- Draw $(X, Y) \stackrel{\text{iid}}{\sim} \text{Unif}(-1, 1)$ and set

$$h(X, Y) = \mathbf{1}\{X^2 + Y^2 \leq 1\}.$$

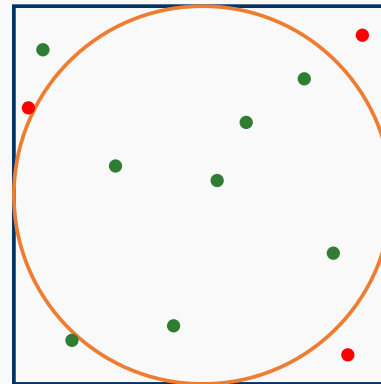


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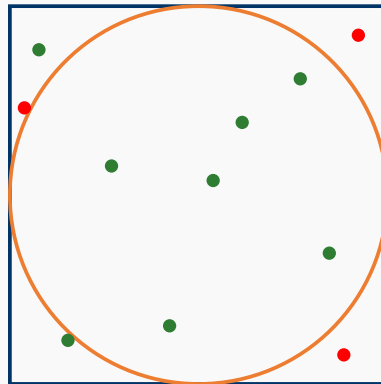


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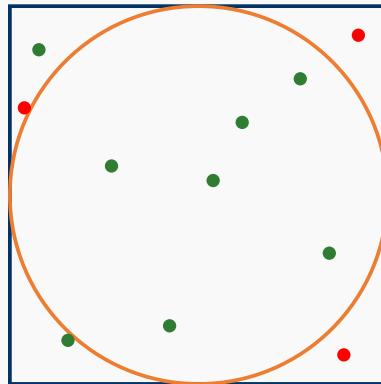


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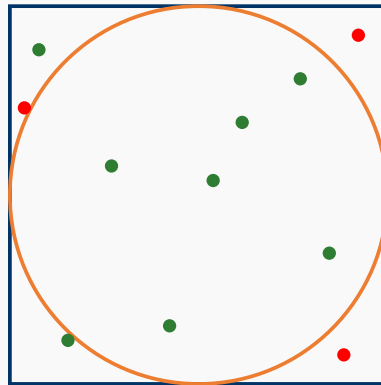


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This is the oldest and most intuitive illustration of the Monte Carlo principle – and it is literally an integral (area) turned into a sample proportion.